

IBVAP — Exact Codebase Map & Module Inventory

03. Comprehensive Inventory & Symbol Mapping

1. Primary Source Files Inventory

File Path	Purpose	Key Functions / Classes	Inputs	Outputs	Deployment Role
src/main.py	Core execution script for YOLO tracking	Loop logic using Ultralytics & OpenCV	../data/video.mp4, yolo11s.pt	GUI Window display	Development / Standalone Pipeline
src/app.py	Complete single-file Streamlit web app	fetch_events, _try_mongodb, _try_csv, verify_secure_audit_log, create_blockchain_ledger, verify_blockchain_ledger, render_*	Env Vars, CSVs, JSON, Videos	Full Interactive Web App	Production / Streamlit Cloud Option 1
streamlit_app/app.py	Modular Streamlit entry point	main(), navigate(), render_sidebar(), render_header()	Session State, Data Service	Navigated Web View	Production / Streamlit Cloud Option 2
streamlit_app/utils/data_service.py	Abstraction layer for event data	IntrusionEvent, fetch_events, _try_supabase, _try_csv, _seed_events	Supabase API, CSV file	List of IntrusionEvent objects	Production Utility
streamlit_app/components/ui_components.py	Custom HTML/CSS component generators	risk_badge, status_indicator, metric_card_html, security_pipeline_html	Component parameters	HTML formatted string	Production UI Component
streamlit_app/pages/command_center.py	Command Center view renderer	render(events)	Event list	Executive Dashboard UI	Production Page
streamlit_app/pages/surveillance.py	Video Feed & Zone view renderer	render(events)	Event list, Video File	Surveillance Player & Timeline UI	Production Page

<code>streamlit_app/pages/security_events.py</code>	Security Event Management renderer	<code>render(events), _render_event_modal(event)</code>	Event list	Data Table & Modal UI	Production Page
<code>streamlit_app/pages/digital_evidence.py</code>	Evidence Gallery renderer	<code>render(events)</code>	Event list, Image files/URLs	Evidence Grid & Viewer UI	Production Page
<code>streamlit_app/pages/audit_log.py</code>	Audit Log & Integrity renderer	<code>render(events)</code>	Event list, Audit files	Audit Table & Export UI	Production Page
<code>streamlit_app/pages/system_page.py</code>	Technical System info renderer	<code>render(events)</code>	Event list	System Module Info UI	Production Page
<code>bytetrack_day3.yaml</code>	ByteTrack tracker hyperparameters	<code>tracker_type, track_high_thresh, track_buffer, etc.</code>	YAML parser	Tracker Config Dict	Model Config File

2. Configuration & Data Artifacts

Artifact Name	Path	Description / Contents
<code>video.mp4</code>	<code>data/video.mp4</code>	Raw surveillance video input feed (2.19 MB)
<code>yolo11s.pt</code>	<code>models/yolo11s.pt</code>	Pretrained YOLO11 Small PyTorch weights file (19.3 MB)
<code>yolo11n.pt</code>	<code>models/yolo11n.pt</code>	Pretrained YOLO11 Nano PyTorch weights file (5.61 MB)
<code>intrusion_log.csv</code>	<code>output/intrusion_log.csv</code>	Standard event log (Event, Person ID, Frame, Timestamp, Risk)
<code>secure_audit_log.csv</code>	<code>output/secure_audit_log.csv</code>	Cryptographically chained audit log containing previous_hash and record_hash
<code>audit_root_hash.txt</code>	<code>output/audit_root_hash.txt</code>	64-character SHA-256 root digest of the entire audit chain
<code>blockchain_ledger.json</code>	<code>output/blockchain_ledger.json</code>	JSON structure representing Genesis and event blocks linked by SHA-256 digests
<code>day3_zone_events.csv</code>	<code>output/day3_zone_events.csv</code>	Zone transition log recording INITIAL_INSIDE, ENTER, and EXIT events

<code>day3_id_switch_results.csv</code>	<code>output/day3_id_switch_results.csv</code>	Tracking ID stability evaluation log recording ID switches, frame gaps, and spatial distances
<code>day3_id_consistency.csv</code>	<code>output/day3_id_consistency.csv</code>	Track consistency IoU evaluation log across re-identification frames
<code>config.toml</code>	<code>streamlit_app/.streamlit/config.toml</code>	Streamlit dark navy theme color scheme definitions

3. Feature-to-Code Map

Feature	Primary File	Primary Function / Symbol	Input Data	Output / Effect
Video Ingestion	<code>src/main.py</code>	<code>cv2.VideoCapture</code>	Video file path	Raw video frames
Person Detection	<code>src/main.py</code>	<code>model.track(..., classes=[0])</code>	Frame array	Bounding boxes & confidence
Object Tracking	<code>bytetrack_day3.yaml</code>	ByteTrack engine	Detections + history	Track IDs
MongoDB Sync	<code>src/app.py</code>	<code>save_events_to_mongodb()</code>	List of <code>IntrusionEvent</code>	Upserted Mongo documents
MongoDB Fetch	<code>src/app.py</code>	<code>_try_mongodb()</code>	Mongo collection	Parsed event objects
CSV Fallback	<code>src/app.py</code>	<code>_try_csv()</code>	<code>intrusion_log.csv</code>	Parsed event objects
Audit Verification	<code>src/app.py</code>	<code>verify_secure_audit_log()</code>	<code>secure_audit_log.csv</code> , <code>audit_root_hash.txt</code>	Integrity boolean + message
Blockchain Generator	<code>src/app.py</code>	<code>create_blockchain_ledger()</code>	<code>secure_audit_log.csv</code>	Writes <code>blockchain_ledger.json</code>
Blockchain Verification	<code>src/app.py</code>	<code>verify_blockchain_ledger()</code>	<code>blockchain_ledger.json</code>	Integrity boolean + block count
Evidence Fetcher	<code>src/app.py</code>	<code>fetch_image_bytes()</code>	File path or HTTP URL	Binary Image bytes
UI Metric Cards	<code>streamlit_app/components/ui_components.py</code>	<code>metric_card_html()</code>	Label, Value, Icon	Formatted HTML string